

$$K_R = A_R + B_R \text{Re}^{-C_R}$$

where K_F and K_R are the forward and reverse form loss coefficient. A_F and A_R are the Words 4 and 5 of cards CCCN101-N109. Re is the Reynolds number based on mixture fluid properties. If these cards are being used for the form loss calculation, then enter all four words for the appropriate expression. See the discussion regarding the flow area basis assumed for A_F and A_R ; the same information applies for B_F and B_R .

W1(R) $B_F (\geq 0)$. This quantity must be greater than or equal to 0.

W2(R) $C_F (\geq 0)$. This quantity must be greater than or equal to 0.

W3(R) $B_R (\geq 0)$. This quantity must be greater than or equal to 0.

W4(R) $C_R (\geq 0)$. This quantity must be greater than or equal to 0.

8.10.11 Branch Junction Initial Conditions, CCCN201

These cards are required depending on the value of n_j as described for cards CCCN101-N109. The values of N should follow the same approach as used in cards CCCN101-N109.

W1(R) Initial liquid or mass flow (velocity in m/s, ft/s or mass flow in kg/s, lb/s).

W2(R) Initial vapor velocity or mass flow (velocity in m/s, ft/s or mass flow in kg/s, lb/s).

W3(R) Interface velocity (m/s, ft/s). Enter 0.

8.11 Separator Component, SEPARATR

A steam separator component is indicated by SEPARATR for Word 2 on card CCC0000. The connection code is CCCVV000N, where CCC is the component number, VV is the volume number, and N is the face number. For major edits, minor edits, and plot variables, the volume is numbered as CCC010000. The junctions associated with the separator component are numbered as CCCJJ0000, where JJ is the junction number ($01 \leq JJ \leq 03$).

A separator component is a specialized branch component having three junctions. The number of junctions n_j defined below must be 3, and no junctions may connect to this component. The variable N defined below must have values of 1, 2, and 3. For the junctions, $N = 1$ is the vapor outlet junction, $N = 2$ is the liquid fall back junction, and $N = 3$ is the separator inlet junction. The from part of the vapor outlet junction must refer to the outlet of the separator (CCC010002), and the from part of the liquid fall back must refer to the inlet of the separator (CCC010001). To include the direct path from a steam generator downcomer to the steam dome, a bypass volume is recommended. The smooth or abrupt junction option

can be used for the three junctions. Appropriate user-input energy loss coefficients may be needed to match a known pressure drop across the separator. We recommend that choking be turned off for all three junctions. The vapor outlet and liquid fall back junctions should use the nonhomogeneous option.

8.11.1 Separator Component Information, CCC0001

This card is required.

- W1(I) Number of junctions, nj. The variable nj is the number of junctions described in the input data for this component and must be 3.
- W2(I) Initial condition control. This word is optional and, if missing, the junction initial velocities in the first and second words on card CCCN201 are assumed to be velocities. If 0, velocities are assumed; if nonzero, mass flows are assumed.

8.11.2 Separator Options, CCC0002

This card is optional. The first word specifies the separator option while the second word specifies the number of actual separator components represented by this RELAP5 SEPARATOR component. The second word is needed if the user uses the General Electric separator options

- W1(I) Separator option, ISEPST. A value of 0 specifies the simple separator contained in previous versions of RELAP5 (default), a value of 1 specifies the General Electric dryer model, a value of 2 specifies a General Electric two-stage separator, and a value of 3 specifies a three-stage General Electric separator.
- W2(I) Number of separator components represented by this RELAP5 component. The number is needed only if Word 1 has a value of 2 or 3.

8.11.3 Separator X-Coordinate Volume Data, CCC0101-0109

This card (or cards) is required. The nine words can be entered on one or more cards, and the card numbers need not be consecutive.

- W1(R) Area (m^2 , ft^2).
- W2(R) Length (m, ft).
- W3(R) Volume (m^3 , ft^3). The code requires that the volume equals the area times the length ($W3 = W1 \bullet W2$). At least two of the three quantities, W1, W2, and W3, must be nonzero. If one of the quantities is 0, it will be computed from the other two. If none of the words are 0, the volume must equal the x-direction area times the x-direction length within a relative error of 0.000001. The same relative error check is done for the y- and z-directions.

- W4(R) Azimuthal angle (degrees). The absolute value of this angle must be ≤ 360 degrees. This quantity is not used in the calculation but is specified for possible automated drawing of nodalization diagrams.
- W5(R) Inclination angle (degrees). The absolute value of this angle must be ≤ 90 degrees. The angle 0 degrees is horizontal, and positive angles have an upward inclination, i.e., the inlet is at the lowest elevation. This angle is used in the interphase drag calculation.
- W6(R) Elevation change (m, ft). A positive value is an increase in elevation. The absolute value of this quantity must be less than or equal to the length. If the vertical angle orientation is 0, this quantity must be 0. If the vertical angle is nonzero, this quantity must also be nonzero and have the same sign. When the absolute value of the elevation angle determined by the ratio of the elevation change (this Word 6) and the length (Word 2) is less than or equal to 45 degrees, the horizontal flow regime map is used. When the ratio is greater than 45 degrees, the vertical flow regime map is used.
- W7(R) Wall roughness (m, ft).
- W8(R) Hydraulic diameter (m, ft). This should be computed from $4.0 \cdot \left(\frac{\text{area}}{\text{wetted perimeter}} \right)$.
If 0, the hydraulic diameter is computed from $2.0 \cdot \left(\frac{\text{area}}{\pi} \right)^{0.5}$. A check is made that the pipe roughness is less than half the hydraulic diameter. See Word 1 for the area.
- W9(I) Volume control flags. This word has the format tlpvbfe. It is not necessary to input leading zeros. Volume flags consist of scalar oriented and coordinate direction oriented flags. Only one value for a scalar oriented flag is entered per volume but up to three coordinate oriented flags can be entered for a volume, one for each coordinate direction. At present, the f flag is the only coordinate direction oriented flag. This word enters the scalar oriented flags and the x-coordinate flag. For the SEPARATR component, the volume control flag is limited to the format 000001e.
- e-flag The digit e specifies if nonequilibrium or equilibrium is to be used. e = 0 specifies that a nonequilibrium (unequal temperature) calculation is to be used, and e = 1 specifies that an equilibrium (equal temperature) calculation is to be used. Equilibrium volumes should not be connected to nonequilibrium volumes. The equilibrium option is provided only for comparison to other codes.

8.11.4 Separator Additional Wall Friction, CCC0131

This card is optional. If this card is not entered, the default values are 1.0 for the laminar shape factor and 0.0 for the viscosity ratio exponent. Two, four, or six quantities may be entered on the card, and the data not entered are set to default values.

W1(R)	Shape factor for x-coordinate.
W2(R)	Viscosity ratio exponent for x-coordinate.
W3(R)	Shape factor for y-coordinate.
W4(R)	Viscosity ratio exponent for y-coordinate.
W5(R)	Shape factor for z-coordinate.
W6(R)	Viscosity ratio exponent for z-coordinate.

8.11.5 Separator Volume Initial Conditions, CCC0200

This card is required.

W1(I)	Control word. This word has the format $\underline{\epsilon}\underline{b}\underline{t}$. It is not necessary to input leading zeros.
$\underline{\epsilon}$ -flag	The digit $\underline{\epsilon}$ specifies the fluid, where $\underline{\epsilon} = 0$ is the default fluid. The value for $\underline{\epsilon} > 0$ corresponds to the position number of the fluid type indicated on the 120-129 cards (i.e., $\underline{\epsilon} = 1$ specifies H_2O , $\underline{\epsilon} = 2$ specifies D_2O etc.). The default fluid is that set for the hydrodynamic system by cards 120-129 or this control word in another volume in this hydrodynamic system. The fluid type set on cards 120-129 or these control words must be consistent (i.e., not specify different fluids). If cards 120-129 are not entered and all control words use the default $\underline{\epsilon} = 0$, then water is assumed to be the fluid. It is recommended that the user use the 120-129 cards to set the fluid type in the systems.
b-flag	The digit $\underline{b}$ specifies whether boron is present. $\underline{b} = 0$ specifies that the volume liquid does not contain boron, and $\underline{b} = 1$ specifies that a boron concentration in mass of boron per mass of liquid (which may be 0) is being entered after the other required thermodynamic information.
t-flag	The digit $\underline{t}$ specifies how the following words are to be used to determine the initial thermodynamic state. $\underline{t} = 0-3$ specifies one component (steam/water); $\underline{t} = 4-6$ allows the specification of two components (steam/water and noncondensable gas).

With options $\underline{t}$ equal to 4-6, names of the components of the noncondensable gas must be entered on card 110, and mass fractions of the noncondensable gas components are entered on card 115.

One Component (Steam/Water), Equilibrium or Non-equilibrium

[P,U_f,U_g,α_g] If $\underline{t} = 0$, the next four words are interpreted as pressure (Pa, lb_f/in²), liquid specific internal energy (J/kg, Btu/lb), vapor specific internal energy (J/kg, Btu/lb), and vapor void fraction. These quantities will be interpreted as nonequilibrium or equilibrium conditions depending on the internal energies used to define the thermodynamic state. Enter boron in W6 if needed.

One Component (Steam/Water), Equilibrium

[T,x_s] If $\underline{t} = 1$, the next two words are interpreted as temperature (K, °F) and static quality in equilibrium condition. Enter boron in W4 if needed.

[P,x_s] If $\underline{t} = 2$, the next two words are interpreted as pressure (Pa, lb_f/in²) and static quality in equilibrium condition. Enter boron in W4 if needed.

[P,T] If $\underline{t} = 3$, the next two words are interpreted as pressure (Pa, lb_f/in²) and temperature (K, °F) in equilibrium condition. Enter boron in W4 if needed.

Two Components (Steam/Water/Air), Non-equilibrium

The following options are used for input of non-equilibrium two-phase and/or noncondensable states. In all cases, the criteria used for determining the range of values for static quality (x_s) are

Two-phase if $1.0\text{E-}9 \leq x_s \leq 0.99999999$,

Single phase if $x_s < 1.0 \text{ E-}9$ or $x_s > 0.99999999$

The static quality is given by $x_s = M_g / (M_g + M_f)$, where $M_g = M_s + M_n$.

[P,T,x_s] If $\underline{t} = 4$, the next three words are interpreted as pressure (Pa, lb_f/in²), temperature (K, °F), and static quality in equilibrium condition. Using this input option with static quality greater than 0.0 and less than or equal to 1.0, saturated noncondensables will result. Also, the temperature is restricted to be less than the saturation temperature at the input pressure. Setting static quality to 0.0 is used as a flag that will initialize the volume to all noncondensable (dry noncondensable) with no temperature restrictions. Static quality is reset to 1.0 using this dry noncondensable option. Enter boron in W5 if needed.

[T,x_s,x_n] If $\underline{t} = 5$, the next three words are interpreted as steam saturation temperature (K, °F), static quality, and noncondensable quality in equilibrium condition. Enter boron in W5 if needed. Both the static and noncondensable qualities are restricted to be between 1.0 E-9 and 0.99999999. The liquid will be subcooled. The input temperature determines the

steam partial pressure, and the total pressure may be greater than the critical pressure, which will cause an input processing error. Little experience has been obtained using this option, and it has not been checked out.

[P,U_f,U_g, α_g ,x_n] If $\underline{t} = 6$, the next five words are interpreted as pressure (Pa, lb_f/in.²), liquid specific internal energy (J/kg, Btu/lb), vapor specific internal energy (J/kg, Btu/lb), vapor void fraction, and noncondensable quality. These quantities will be interpreted as nonequilibrium or equilibrium conditions depending on the internal energies used to define the thermodynamic state. The combinations of vapor void fraction and noncondensable quality must be thermodynamically consistent. If noncondensable quality is set to 0.0, noncondensables are not present and the input processing branches to that type of processing ($\underline{t} = 0$). If noncondensables are present (noncondensable quality greater than 0.0), then the vapor void fraction must also be greater than 0.0. If the noncondensable quality is set to 1.0 (pure noncondensable), then vapor void fraction must also be 1.0. When both the vapor void fraction and the noncondensable quality are set to 1.0, the volume temperature is calculated from the noncondensable energy equation using the input vapor-specific internal energy. Enter boron in W7 if needed.

W2-W7(R) Quantities as described under Word 1. Depending on the control word, 2-6 quantities may be required. Enter only the minimum number required. If entered, boron concentration (mass of boron per mass of liquid) follows the last required word for thermodynamic conditions.

8.11.6 Separator Junction Geometry, CCCN101-N109

These cards are required. Cards with N equal to 1, 2 and 3 are entered, one for each junction.

- W1(I) From connection code to a component. This refers to the component from which the junction coordinate direction originates. The connection code is CCCVV000N, where CCC is the component number, VV is the volume number, and N indicates the face number. The number N equal to 1 and 2 specifies the inlet and outlet faces respectively for the volume's coordinate direction. The number N equal to 3-6 specifies crossflow. The number N equal to 3 and 4 would specify inlet and outlet faces for the second coordinate direction; N equal to 5 and 6 would do the same for the third coordinate direction. For connecting to a time-dependent volume, N is restricted to 1 or 2.
- W2(I) To connection code to a component. This refers to the component at which the junction coordinate direction ends. See the description for W1 above.
- W3(R) Area (m², ft²). If 0, the area is set to the minimum area of the adjoining volumes. For abrupt area changes, the junction area must be equal to or smaller than the minimum of the adjoining areas. For smooth area changes, there are no restrictions.

- W4(R) Reynolds number independent forward flow energy loss coefficient, A_F . This quantity will be used in each of the phasic momentum equations when the junction velocity of that phase is positive or 0. A variable loss coefficient may be specified (see Section 8.10.10). The interpretation and use of this loss coefficient vary based upon the “a” junction control flag selected in W6. If the smooth area change option is selected ($a = 0$), the entry here is the loss coefficient appropriate for the junction flow area resulting from the entry made in W3. If one of the abrupt area change options is selected ($a = 1$ or 2), the entry here is the loss coefficient appropriate for the minimum flow area between the two hydrodynamic volumes connected by this junction. If the $a = 1$ option is selected, a loss coefficient entered here is considered to be additive with the abrupt area change flow energy loss coefficient, which is calculated separately by the code.
- W5(R) Reynolds number independent reverse flow energy loss coefficient, A_R . This quantity will be used in each of the phasic momentum equations when the junction velocity of that phase is negative. A variable loss coefficient may be specified (see Section 8.10.10). See information for the previous word regarding interpretation and use of this loss coefficient.
- W6(I) Junction control flags. This word has the format jefvcahs. For this component, the junction control flag format is limited to 0000cahs.
- c-flag The digit c specifies choking options. c = 0 means that the choking model will be applied, and c = 1 means that the choking model will not be applied. The choking model used is based on input on card 1. If card 1 is missing or it does not contain Option 50, the Henry-Fauske critical flow model is active. If card 1 is present and it contains Option 50, the original RELAP5 critical flow model is active.
- a-flag The digit a specifies area change options. a = 0 means a smooth area change and only the user supplied K_{loss} is applied. a = 1 means full abrupt area change model and the code calculates forward and reverse contraction/expansion K_{loss} terms and adds them to the user supplied K_{loss} terms. In addition, the code includes extra interphase drag to account for the vena contracta. a = 2 means a partial abrupt area change model and it is the same as “a = 1 except no code calculated forward and reverse contraction/expansion K_{loss} is applied.
- h-flag The digit h specifies nonhomogeneous or homogeneous. h = 0 specifies the nonhomogeneous (two-velocity momentum equations) option and h = 1 or 2 specifies the homogeneous (single-velocity momentum equation) option. For the homogeneous option (h = 1 or 2), the major edit printout will show h = 1.
- s-flag The digit s specifies momentum flux options. s = 0 uses momentum flux in both the to and the from volume. s = 1 uses momentum flux in the from volume, but not in the to volume.

$\underline{s} = 2$ uses momentum flux in the to volume, but not in the from volume. $\underline{s} = 3$ does not use momentum flux in either the to volume or the from volume.

W7(R) Void fraction limit. For the vapor exit junction ($N = 1$), this quantity (VOVER) is the vapor void fraction above which flow out of the vapor outlet is pure vapor. If the word is missing, a default value of 0.5 is used. For the liquid fall back junction ($N = 2$), this quantity (VUNDER) is the liquid void fraction above which flow out of the liquid fall back is pure liquid. If the word is missing, a default value of 0.15 is used. For the separator inlet, this word is not used.

8.11.7 Separator Junction Form Loss Data, CCCN112

These cards are optional. The values of N should follow the same approach as used in cards CCCN101-N109. The user-specified form loss is given in Words 4 and 5 of cards CCCN101-N109 if these cards are not entered. If these cards are entered, the form loss coefficient is calculated from

$$K_F = A_F + B_F Re^{-C_F}$$

$$K_R = A_R + B_R Re^{-C_R}$$

where K_F and K_R are the forward and reverse form loss coefficient. A_F and A_R are the Words 4 and 5 of cards CCCN101-N109. Re is the Reynolds number based on mixture fluid properties. If these cards are being used for the form loss calculation, then enter all four words for the appropriate expression. See the discussion regarding the flow area basis assumed for A_F and A_R ; the same information applies for B_F and B_R .

W1(R) $B_F (\geq 0)$. This quantity must be greater than or equal to 0.

W2(R) $C_F (\geq 0)$. This quantity must be greater than or equal to 0.

W3(R) $B_R (\geq 0)$. This quantity must be greater than or equal to 0.

W4(R) $C_R (\geq 0)$. This quantity must be greater than or equal to 0.

8.11.8 Separator Junction Initial Conditions, CCCN201

These cards are required depending on the value of n_j as described for cards CCCN101-N109. The values of N should follow the same approach as used in cards CCCN101-N109. A 90% extraction limit during input processing is tested for the vapor at the vapor outlet junction and for the liquid fall back junction. If greater than 90%, an input error occurs.

W1(R) Initial liquid or mass flow (velocity in m/s, ft/s or mass flow in kg/s, lb/s).

W2(R) Initial vapor velocity or mass flow (velocity in m/s, ft/s or mass flow in kg/s, lb/s).

W3(R) Interface velocity (m/s, ft/s). Enter 0.

8.11.9 GE Separator Data, CCC0500

This card is optional for the GE separator. If this card is missing and the GE separator has been specified on card CCC0002, the default values will be used. If the card is present, all eight values must be specified.

W1(R) Radius of larger pickoff ring at first stage of a two-stage separator (m, ft). (Default = 0.0857208 m).

W2(R) Standpipe flow area (m^2 , ft^2). Default = 0.018637 m^2 .

W3(R) Separator nozzle exit area (m^2 , ft^2). Default = 0.01441 m^2 .

W4(R) Radius of separator hub at inlet (m, ft). Default = 0.0809585 m.

W5(R) Swirl vane angle relative to the horizontal (deg). Default = 48 deg.

W6(R) Liquid carryover coefficient for upper separating stages. Default = 0.009 for two-stage separator and 0.110 for three-stage separator.

W7(R) Vapor carryunder coefficient for upper separating stages. Default = 0.0004.

W8(R) Axial distance between exit of first stage discharge passage and swirl vanes (m, ft). Default = 0.2127 m for two-stage separator and 0.45083 m for three-stage separator.

8.11.10 GE Separator First Stage Data, CCC0501

This card is optional for the GE separator. If this card is missing and the GE separator has been specified on card CCC0002, the default values will be used. If the card is present, all nine values must be specified.

W1(R) Liquid film void profile coefficient. Default = 110.0.

W2(R) Vapor core void profile coefficient. Default = 0.5.

W3(R) Separator wall inner radius (m, ft). Default = 0.10794 m.

W4(R) Pickoff ring inner radius (m, ft). Default = 0.069875 m for two-stage separator and 0.0857208 m for three-stage separator.

W5(R)	Discharge passage flow area (m^2 , ft^2). Default = 0.0415776 m^2 for two-stage separator and 0.0096265 m^2 for three-stage separator.
W6(R)	Discharge passage hydraulic diameter (m, ft). Default = 0.045558 m for two-stage separator and 0.025399 m for three-stage separator.
W7(R)	Separating barrel length (m, ft). Default = 0.877845 m for two-stage separator and 1.0699 m for three-stage separator.
W8(R)	Discharge passage loss coefficient. Default = 10.0 for two-stage separator and 2.5 for three-stage separator.
W9(R)	Discharge passage effective L/D coefficient. Default = 450.0 for two-stage separator and 53.44 for three-stage separator.

8.11.11 GE Separator Second Stage Data, CCC0502

This card is optional for the GE separator. If this card is missing and the GE separator has been specified on card CCC0002, the default values will be used. If the card is present, all nine values must be specified.

W1(R)	Liquid film void profile coefficient. Default = 20.0.
W2(R)	Vapor core void profile coefficient. Default = 0.25.
W3(R)	Separator wall inner radius (m, ft). Default = 0.06985 m for two-stage separator and 0.10794 m for three-stage separator.
W4(R)	Pickoff ring inner radius (m, ft). Default = 0.06032 m for two-stage separator and 0.0952453 m for three-stage separator.
W5(R)	Discharge passage flow area (m^2 , ft^2). Default = 0.0029133 m^2 for two-stage separator and 0.0096265 m^2 for three-stage separator.
W6(R)	Discharge passage hydraulic diameter (m, ft). Default = 0.0121699 m for two-stage separator and 0.025399 m for three-stage separator.
W7(R)	Separating barrel length (m, ft). Default = 0.16255 m for two-stage separator and 0.384156 m for three-stage separator.
W8(R)	Discharge passage loss coefficient. Default = 0.5 for two-stage separator and 1.429 for three-stage separator.

W9(R) Discharge passage effective $\frac{L}{D}$ coefficient. Default = 95.85 for two-stage separator and 194.64 for three-stage separator.

8.11.12 GE Separator Third Stage Data, CCC0503

This card is optional for the GE separator. If this card is missing and the GE three-stage separator has been specified on card CCC0002, the default values will be used. If the card is present, all nine values must be specified.

W1(R) Liquid film void profile coefficient. Default = 20.0.

W2(R) Vapor core void profile coefficient. Default = 0.55.

W3(R) Separator wall inner radius (m, ft). Default = 0.10794 m.

W4(R) Pickoff ring inner radius (m, ft). Default = 0.0984201 m.

W5(R) Discharge passage flow area (m², ft²). Default = 0.0096265 m².

W6(R) Discharge passage hydraulic diameter (m, ft). Default = 0.025399 m.

W7(R) Separating barrel length (m, ft). Default = 0.384156 m.

W8(R) Discharge passage loss coefficient. Default = 2.563.

W9(R) Discharge passage effective L/D coefficient. Default = 424.96.

8.11.13 GE Dryer Data, CCC0600

This card is optional for the GE separator. If this card is missing and the GE dryer has been specified on card CCC0002, the default values will be used. If the card is present, all three values must be specified.

W1(R) Vapor velocity at dryer inlet below which there is 0% liquid carryover (m/s, ft/s). Default = 1.5 m/s.

W2(R) Vapor velocity at dryer inlet above which there is 100% liquid carryover (m/s, ft/s). Default = 6.0 m/s.

W3(R) Range of dryer inlet quality where dryer carryover changes from 0 to 100% when dryer inlet vapor velocity is between lower and upper values. Default = 0.05.